Research Square

Preprints are preliminary reports that have not undergone peer review. They should not be considered conclusive, used to inform clinical practice, or referenced by the media as validated information.

ML-Driven Quantum Portfolio Optimization: Hybrid CNN-LSTM Architecture with Adaptive Zero-Noise Extrapolation on NISQ Devices

Maryam Arif

ranamaryamarif@gmail.com

COMSATS University Islamabad, Pakistan https://orcid.org/0000-0003-3047-3012

Soban Saeed

National University of Sciences & Technology (NUST), Islamabad, Pakistan https://orcid.org/0009-0006-7279-9714

Research Article

Keywords: Quantum Portfolio Optimization, NISQ Devices, Machine Learning-based Zero Noise Extrapolation (ML-ZNE), Hybrid CNN-LSTM Model, Quantum Error Mitigation, Quantum Finance

Posted Date: December 25th, 2025

DOI: https://doi.org/10.21203/rs.3.rs-8423464/v1

License: This work is licensed under a Creative Commons Attribution 4.0 International License. Read Full License

Additional Declarations: The authors declare no competing interests.

ML-Driven Quantum Portfolio Optimization: Hybrid CNN-LSTM Architecture with Adaptive Zero-Noise Extrapolation on NISQ Devices

Maryam Arif *, Soban Saeed†
*Department of Management Sciences, COMSATS University Islamabad, Pakistan
Email: fa21-bba-095@isbstudent.comsats.edu.pk
†Department of Electrical Engineering, National University of Sciences & Technology (NUST), Islamabad, Pakistan
Email: ssaeed.bee2021mcs@student.nust.edu.pk

***Abstract*—The quantum portfolio optimization can be hindered by intrinsic hardware imperfections for noisy intermediate-scale quantum (NISQ), which can impair the precision of expectation value calculations as well as constrain financial outcomes. Our study proposes a machine learning-based zero noise extrapolation (ML-ZNE) approach that can employ a hybrid convolutional neural network and long short-term memory (CNN-LSTM) structure, which can also have an adaptive noise scaling to reduce these errors. The method dynamically estimates the noise-free expectation values from a series of noise-scaled measurements and substitutes traditional static extrapolation methods with a data-driven framework. The CNN-LSTM model can identify the spatial and temporal relationships in noisy quantum circuit outputs that can result in more robust extrapolation than polynomial-fitting-based ZNE. Moreover, the approach includes the real-time hardware calibration data to refine the noise scaling parameters that can elevate the accuracy of quantum computations. Our experimental validation shows that ML-ZNE can decrease both the extrapolation error and improve the portfolio performance, which can quantify the Sharpe ratio while achieving better results than both static ZNE and unmitigated optimization. The proposed method improves the utility of real-world quantum finance by tackling key noise-induced constraints without compromising processing speed on NISQ devices. This work connects the quantum error correction alongside machine learning and presents a scalable approach for advanced quantum portfolio management.**

***Index Terms*—Quantum Portfolio Optimization, NISQ Devices, Machine Learning-based Zero Noise Extrapolation (ML-ZNE), Hybrid CNN-LSTM Model, Quantum Error Mitigation, Quantum Finance**

I. Introduction

QUANTUM computing can be considered a transformative technology as it contains significant potential in financial engineering, including the portfolio optimization problems, which require complex combinatorial calculations [8]. Quantum approaches can be utilized to evaluate the ability of multiple asset combinations simultaneously [14]. Nevertheless, current noisy intermediate-scale quantum (NISQ) devices face challenges due to limited qubit coherence times as well as inherent noise, which impacts computational accuracy [25]. To mitigate these noise effects in quantum computations, zero-noise extrapolation (ZNE) has become an essential technique [13]. Traditionally, ZNE implementations typically employ static noise factors that can cause the failure to account for the dynamic nature of quantum hardware noise characteristics. The limitation in ZNE becomes particularly problematic in calculations where precision is crucial for reliable decision-making. [28]. The noise profile of quantum devices evolves due to environmental factors and calibration drift, which not only varies between different hardware platforms [3].

Promising approaches to address multiple challenges in quantum computing have been demonstrated for recent advances in machine learning [16]. Particular success is observed in processing complex, multi-dimensional data patterns from hybrid architectures that combine convolutional neural networks (CNNs) and long short-term memory (LSTM) networks. [23].

CNNs extract spatial features from structured data, excelling in comparison to LSTMs, which effectively model temporal dependencies in sequential information [18]. Such characteristics make them well-suited for analyzing quantum hardware calibration data, which contains spatial correlations across qubits as well as temporal noise variations. Our study presents a novel approach that integrates a hybrid CNN-LSTM model with ZNE for optimization of the quantum portfolio.

The method predicts optimal noise scaling factors dynamically based on real-time quantum hardware calibration data, which can address the limitations of static ZNE implementations. Our study approach differs from existing methods in three aspects: Firstly, rather than relying on fixed parameters, our study continuously adapts to changing noise conditions. Secondly, through hybrid architecture, it considers both spatial as well as temporal noise characteristics. Lastly, it targets with a focus on the requirements of financial applications where accurate expectation values are crucial for a reliable portfolio optimization [31].

Our proposed methods contribute to the field through a more robust and adaptive framework to mitigate errors in quantum financial computing. We combine machine learning with quantum error mitigation techniques to bridge the gap between theoretical quantum advantage as well as practical implementation on current NISQ devices. This approach is valuable to explore quantum computing applications benefiting

financial institutions, as it offers better reliability in portfolio optimization tasks [1].

II. Literature Review

With the progress in quantum computing over recent years, especially in portfolio problem optimization, where instead of classical methods, promising quantum algorithms are becoming promising for potential speedups. For the combinatorial optimization problem approach, the quantum approximate optimization algorithm (QAOA) has become an emerging new approach (Farhi et al., 2014) [11]. Still, for practical implementation on NISQ devices, they face a significant challenge due to both noise and limited qubit coherence times, which can facilitate advanced error mitigation techniques.

A. Quantum Portfolio Optimization

Quantum algorithms can solve problems more efficiently due to modern portfolio optimization, which builds on Markowitz's mean-variance framework (Fabozzi et al., 2008) [9]. Various quantum approaches, such as quantum annealing (Venturelli & Kondratyev, 2019) [27] and gate-model quantum algorithms (Buonaiuto et al., 2023) have been explored [4]. The work demonstrated practical implementations using the variational quantum eigen solvers by real quantum devices, although highlighting the crucial need for error mitigation. Recent studies have shown how quantum algorithms can reduce the systemic risk in financial networks (Aboussalah et al., 2023) [1], thus further motivating robust implementations.

B. Error Mitigation Techniques

Useful results can be extracted from NISQ devices using error mitigation, becoming essential in the field. Zero noise extrapolation (ZNE) showcases one of the most widely used techniques, as circuits are executed at multiple noise levels and results are extrapolated to the zero-noise limit (Giurgica-Tiron et al., 2020) [13]. Various noise scaling methods have been presented, not limited to Patil et al. (2023) [20], and the recent study addressing the challenges of time-correlated noise (Schultz et al., 2022) [22]. The alternative approaches can include probabilistic error cancellation (Van Den Berg et al., 2023) [26] and error mitigation measurements with the distinct advantages and limitations (Khan et al., 2024) [15].

C. Machine Learning for Quantum Error Mitigation

Promising results were shown when the integration of machine learning with quantum error mitigation was used. Recent studies showcased how machine learning can reduce the overhead of quantum error mitigation (Liao et al., 2024) [17], especially for ZNE. From noise characterization (Youssry et al., 2020) [30] to circuit optimization (Fan et al., 2022) [10], various quantum computing challenges have neural networks applied to them. Although the application of error mitigation remains underexplored, hybrid quantum-classical machine learning models have shown particular promise (Pulicharla & Eruvaram, 2023) [21].

D. Quantum Noise Modeling

For effective error mitigation, the noise modeling is crucial. Transfer methods (Chen et al., 2020) [6] and machine learning techniques (Cincio et al., 2021) [7] have been employed in recent approaches to characterize complex noise profiles. The particular challenges that conventional ZNE methods overlook present spatial and temporal correlations in quantum noise (Soare et al., 2014) [24].

III. Theoretical Framework

Our study investigates the role of machine learning-driven zero noise extrapolation (ML-ZNE), which can improve the quantum portfolio optimization for noisy intermediate-scale quantum (NISQ) devices. This framework adapts error mitigation techniques and dynamic noise scaling strategies to reduce extrapolation errors, especially in quantum expectation value estimation, as mentioned in our H1 and H3. This can improve the estimated accuracy, which enhances the stability as well as coverage of quantum optimization algorithms that can, in turn, lead to superior portfolio performance that is measured by the Sharpe ratio (H2). These effective relationships can be influenced by the underlying quantum hardware noise characteristics, which can be defined as gate error rates and decoherence times that can moderate the impact of adaptive methods, which are relative to static approaches.

A. Independent Variables

- Error mitigation approach (ML-ZNE, static ZNE, no mitigation)
- Noise scaling strategy (adaptive vs. fixed)
- Use of a hybrid CNN–LSTM model for noise prediction

B. Mediating Variables

- Accuracy of noise-free expectation value estimation
- Stability and convergence of quantum portfolio optimization

C. Dependent Variables

- Extrapolation error (H1, H3)
- Portfolio performance measured by the Sharpe ratio (H2)

D. Moderating Variable

- Quantum hardware noise characteristics (gate errors, decoherence, measurement noise)

IV. Hypothesis

A. H1: Error Mitigation Effectiveness

The proposed ML-ZNE approach using a hybrid CNN–LSTM architecture achieves significantly lower extrapolation error than conventional static ZNE methods on NISQ devices.

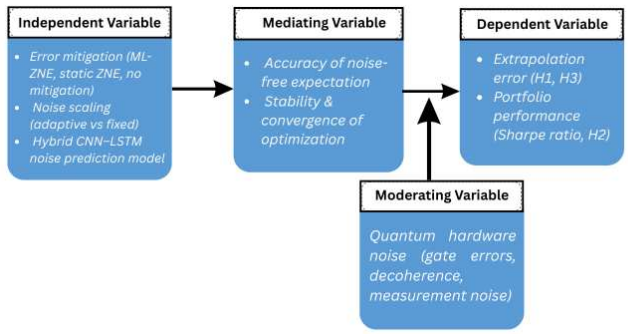

Fig. 1. Theoretical framework showing the impact of error mitigation and noise scaling on optimization and portfolio performance under quantum noise.

B. H2: Financial Performance Improvement

Quantum portfolio optimization employing ML-ZNE yields significantly higher portfolio performance, measured by the Sharpe ratio, compared to optimization using static ZNE techniques or no error mitigation.

C. H3: Benefit of Adaptive Noise Scaling

Dynamic prediction of noise scaling factors using real-time quantum hardware calibration data results in more accurate noise-free expectation value estimation than fixed noise scaling factors.

V. Preliminaries on Zero-Noise Extrapolation and Quantum Portfolio Optimization

A. Zero-Noise Extrapolation Fundamentals

Zero-noise extrapolation (ZNE) follows the principle of intentionally amplifying the quantum circuit noise through control scaling and then extrapolating the results back to the zero noise limit.

For small noise levels, this relationship can be approximated by a low-order polynomial:

$$E(\lambda) = E(0) + \sum_{k=1}^{n} a_k \lambda^k + \mathcal{O}\big(\lambda^{n+1}\big)$$

Where E(0) represents the noiseless expectation values, that we aim to achieve.

The standard implementation involves the three steps:

1) Executing the quantum circuit at multiple noise levels λ_i
2) Measuring corresponding expectation values $E(\lambda_i)$
3) Fitting these points to determine E(0) through Richardson extrapolation [12]

Noise scaling typically involves unitary folding; a technique where circuit layers repeatedly increase the noise level without modifying logical operation. For a circuit with unitary U = $U_L ... U_1$, then the folded version becomes:

$$U(\lambda) = (U_L U_L^+)^k U_L ... (U_1 U_1^+)^k U_1$$

Where k determines the integer noise scale factor $\lambda = 2k+1$ [19].

B. Quantum Portfolio Optimization Framework

The portfolio optimization problem maps to finding the baseline of an Ising Hamiltonian encoding the Markowitz mean-variance model. Given N assets with expected returns μ_i and covariance matrix $\sum_{ij}$, the objective function becomes:

$$H = \sum_{i=1}^{N} \mu_i z_i \; - \; \gamma \sum_{i,j=1}^{N} \Sigma_{ij} \, z_i z_j$$

,

where $z_i \in \{0, 1\}$ represents asset selection and γ controls risk tolerance. The Quantum Approximate Optimization Algorithm (QAOA) prepares a parameterized state

$$|\psi(\vec{\beta}, \vec{\gamma})\rangle = e^{-i\beta_p H_M} e^{-i\gamma_p H_C} \cdots e^{-i\beta_1 H_M} e^{-i\gamma_1 H_C} |+\rangle^{\otimes N},$$

With mixer Hamiltonian $H_M = \sum_{i=1}^{N} X_i$ and cost Hamiltonian H_C, the expectation value $\langle H_C \rangle$ serves as the optimization target, which noise systematically distorts.

C. Noise Characteristics in NISQ Devices

Quantum hardware noise manifest through several mechanisms that affect ZNE performance. Coherent errors arise through imperfect gate calibrations, while incoherent error include:

1) Amplitude damping with time constant T_1
2) Phase damping with time constant T_2
3) Gate errors characterized by Pauli error rates $\in_g$
4) Measurement errors with misassignment probabilities p_m

These noise sources combine to produce a complex non-Markovian dynamics [5] that vary both spatially across qubits and temporally between calibrations [29].

The effective noise scale λ in ZNE depends on:

$$\lambda \propto \exp\left(-\frac{t_g}{T_1}\right) (1 - \epsilon_g)(1 - p_m)$$

Where t_g represents gate duration. This relationship shows that static ZNE approaches often fail because actual noise scaling depends strongly over current device parameters that fluctuate over time [2].

VI. Dataset Description

This study employs two datasets:

1) Quantum generated dataset for ML-ZNE
2) Financial time series data for Portfolio Performance analysis

A. Quantum Dataset for ML-ZNE

The quantum dataset is generated synthetically by adding small perturbations to the feature matrix for each qubit. In total, 100 samples are generated, where each sample comprises calibration information from all 27 qubits of the backend. For each qubit, five noise-related features are extracted, yielding a 27 × 5 feature matrix per sample.

B. Financial Dataset

The Fama-French dataset is utilized for portfolio analysis. The dataset comprises of daily market returns from 02 - 07 - 1990 till 31 - 10 - 2025. The total number of samples are 9223 with 5 features. All the returns obtained from dataset are normalized and used to compute baseline and sharpe ratios for ZNE and ML-ZNE.

VII. Methodology

A. Quantum Noise Modeling

In order to evaluate the validity of the proposed system in realistic noisy intermediate-scale quantum (NISQ) systems, this work implements quantum circuits controlled by a measured 27-qubit falcon processor noise model (FakeMontreal) trained on an IBM Quantum backend. The parameters chosen include qubit relaxation times T_1, T_2 dephasing times, and the error rates of single-qubit (S_X) and two-qubit gates (C_X), as well as measurement readout errors. These extracted parameters are incorporated into a density-matrix simulator, specifically Qiskit Aer Simulator, which allows the simulation framework to realize decoherence and operational perturbations while eliminating finite shot-sampling noise.

Feature	Value
T_1	99.8 μs
T_2	86.1 μs
Average Readout Error	2.5%
S_X	3.5×10^{-4}
C_X	56 Nonzero Couplings

TABLE I
FEATURE VALUES OBTAINED FROM THE FAKEMONTREAL BACKEND

Conversely, noisy references are created through state-vector simulation and used as the ground truth in supervised learning and parameter estimation.

B. Calibration Feature Extraction and Dataset Construction

To enable adaptive noise mitigation, the calibration data is transformed into features. For each qubit, a five dimensional vector with features T_1, T_2, readout error, S_X and C_X concatenated to form a feature vector of 27 x 5.

To simulate temporal drift in feature vectors, synthetic time evolution was introduced by applying small Gaussian perturbations to the feature matrix over ten time steps. This results in a four dimensional tensor of shape (1, 10, 27, 5).

We constructed portfolio optimization instances using historical returns data from Fama–French Factors and Portfolios spanning 1990 - 2025. A six-qubit benchmark circuit was constructed by applying a Hadamard gate followed by a fixed Z-rotation to each qubit, followed by a chain of nearest-neighbor C_X gates. The observable was defined as the average of single-qubit Pauli-Z operators. The implemented quantum circuits were QAOA with a depth of 4, which is optimized using COBYLA classical optimizer.

C. Baseline Methods

We compared ML-ZNE against Linear ZNE using fixed scaling factors [1, 2] with first-order extrapolation.

D. Evaluation Metrics

- Mean absolute error between mitigated and ideal expectation values (H1)
- Comparative accuracy of adaptive versus static extrapolation (H3)
- Deviation of portfolio performance metrics from ideal values (H2)

E. Noise Scaling and Zero-Noise Extrapolation

Let (H)(λ) denote the expectation value of H measured at effective noise scaling factor λ. Noise scaling is implemented by executing modified versions of circuit at multiple noise levels $\lambda \in \lambda_1, \lambda_2, ..., \lambda_k$. Overally, we employed two noise amplification strategies.

1) Simplified Circuit Repetition: This strategy amplifies the noise and avoids imperfect scaling.

2) Standard Unitary Folding: This strategy increases noise strength and preserves ideal circuit evolution.

Static zero noise extrapolation (ZNE) estimates the noise $(H)_0$ by fitting a low-order polynomial, such as:

$$< H > (\lambda) = (H)_0 + c_1\lambda + c_2\lambda^2 + \in,$$

Where c_1 and c_2 are regression coefficients, and $\in$ estimates the higher order noise. In this case, Static ZNE employed second order polynomial fitting across noisy expectation values. While, $(H)_0$ serves as the baseline for comparison with the proposed methodology. Noise scaling factors $\lambda \in 1, 3, 5, 7$ were used depending on the experiment.

F. Machine Learning–Driven Zero-Noise Extrapolation (ML-ZNE)

The ML-ZNE model replaces the fixed extrapolation criteria through an adaptive and learning approach by mapping noisy values with the noiseless counterparts. The following function illustrates the learning:

$$(H)_0 = f_\theta(\{< H > (\lambda_i)\}_{i=1}^{k}, Z$$

Where f_θ is a neural network, $\{(H)(\lambda_i)\}$ are noise scaled expectation values, while z represents optional hardware calibration.

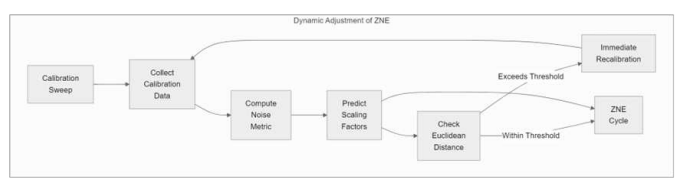

Fig. 2. Dynamic Adjustment of ZNE Process Flow

1) Input Representation: The input to the model involves a sequence of expectation values extracted from increasing noise scales:

$$X = [< H > (\lambda_1), < H > (\lambda_2), ..., < H > (\lambda_k)]$$

2) CNN + LSTM Model: A one-dimensional convolutional layer is applied to the input sequence to extract spatial correlations among noise measurements. The convolution operation is defined as:

$$h_{CNN} = \sigma(W_c * x + b_c)$$

Where W_c and b_c denote convolutional weights and biases, * represents convolution, σ is a nonlinear activation function.

The CNN output is passed to a long short term memory (LSTM) layer, which models sequential correlations across noise scales. The LSTM updates its hidden layer according to:

$$h_t = LSTM(h_{t-1}, h_{CNN,t})$$

This allows the model to learn how expectation values adapts to noise changes.

3) Output and Training Objective: The final hidden state of LSTM is passed to fully connected layer to predict a scalar output:

$$(H)_0 = W_0 h_T + b_0$$

The model is trained in a supervised manner using pairs of noisy and ideal expectation values. The loss function is defined by:

$$L(\theta) = 1/N \sum_{i=1}^{N} |f_\theta(X_i) - (H)_0^{(i)}|$$

The optimization is performed using gradient based methods. The model is implemented in PyTorch and trained using the Adam optimizer with a learning rate of 10^{-4} for 100 epochs.

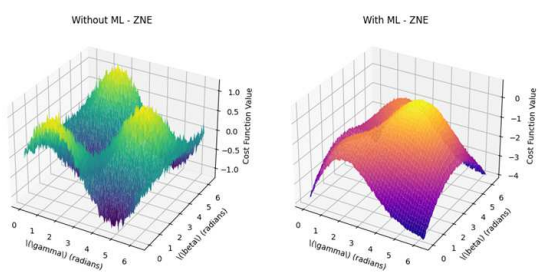

Fig. 3. QAOA cost landscapes for 5-asset portfolio optimization with and without ML-ZNE

4) Quantum Portfolio Optimization Mapping: To assess the impact, the ML-ZNE system is applied to quantum optimization problem. Asset returns and covariances are encoded into quadratic Hamiltonian of the form:

$$H = \sum_i \mu_i x_i - \gamma \sum_{i,j} x_i x_j$$

Where $x_i \in$ 0,1 represent asset selection, μ_i denotes expected returns, $\sum_{ij}$ is the covariance matrix, and γ is a risk aversion parameter.

Expectation values of H are estimated using static ZNE and ML ZNE. Portfolio performance is evaluated using a Sharpe ratio to quantify sensitivity to estimation accuracy.

To evaluate the generalization capability of the proposed ML-ZNE framework, quantum circuits of varying depths were considered. Specifically, the circuit depth parameter was varied as n $\in$ $1, 2, 4, 6$, and the performance of ML-ZNE was compared against conventional static ZNE across these depths.

VIII. Results

A. Noise-Induced Bias and Baseline Performance

For a six-qubit benchmark circuit, the ideal expectations are approximated with zero. Under the realistic noise ($\lambda = 1$), the resulting expectation deviates from zero. This observation confirms that the hardware noise distorts the ideal circuit conditions. Increasing the noise scaling factor results in larger deviations from ideal conditions.

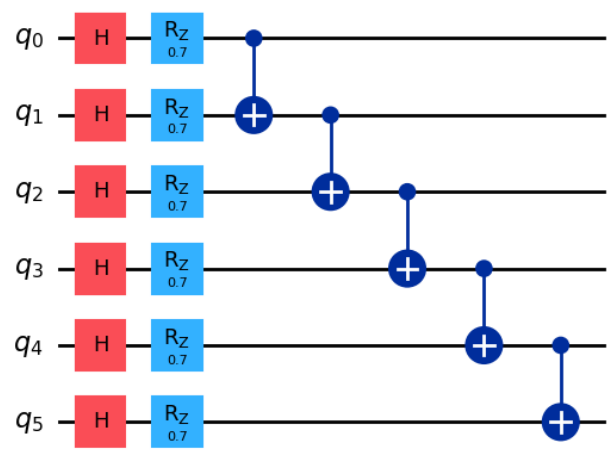

Fig. 4. Circuit Diagram to deviate expectations from ideal situation

B. Static Zero-Noise Extrapolation

Static ZNE did not consistently improve the results. In low noise settings, the linear extrapolation amplifies the error relative to unmitigated results. Polynomial extrapolation illustrates stability for specific circuits but fails to generalize across depths and profiles.

C. Machine-Learning-Based Zero-Noise Extrapolation

The adaptive, calibration-based CNN-LSTM model converged during training and achieved a near-zero loss. This model, when applied to noisy measurements, consistently reduced absolute errors as compared to static ZNE.

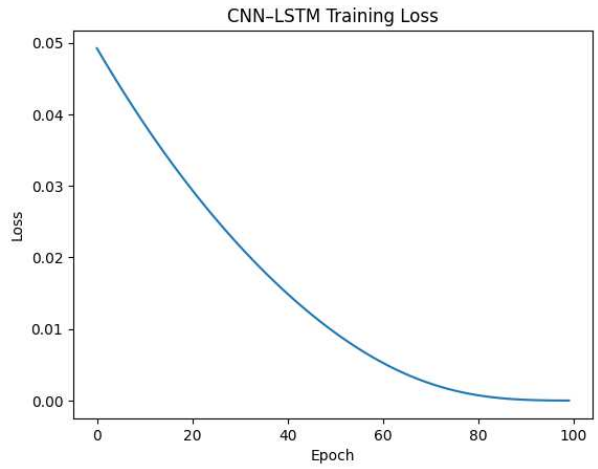

Fig. 5. Loss Training Curves for CNN-LSTM Model

1) H1: Stability of ML-ZNE for Expectation Value Estimation: Hypothesis H1 examines whether ML-ZNE provides more reliable expectation value as compared to static ZNE under realistic NISQ noise.

Figure 6 compares the absolute error for static ZNE and ML-ZNE

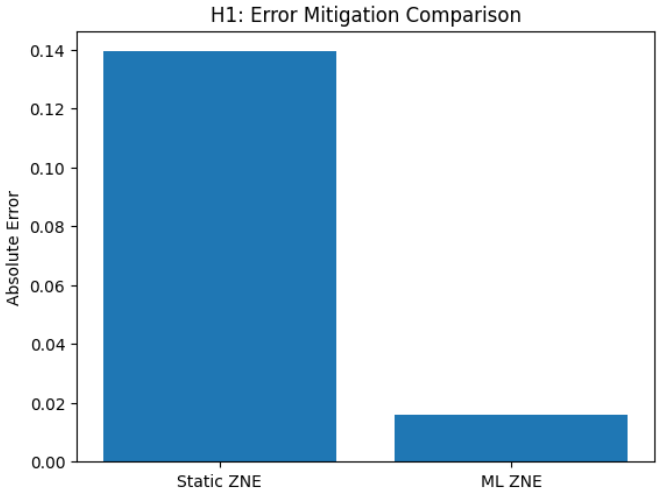

Fig. 6. Absolute error comparison for expectation value estimation

These findings support VIII-C1 H1 in terms of robustness and stability, along with absolute error minimization.

D. Portfolio Optimization Performance

In QAOA portfolio optimization experiment, the ML-ZNE expectation values are closer to the ideal values than the static ZNE. These expectation values, when translated into financial performance, yield a higher Sharpe ratio as compared to static ZNE. This demonstrates improved noise mitigation.

1) H2: Impact on Portfolio Performance: Hypothesis H2 evaluates whether improved expectation value stability translates into improved application-level performance in quantum portfolio optimization.

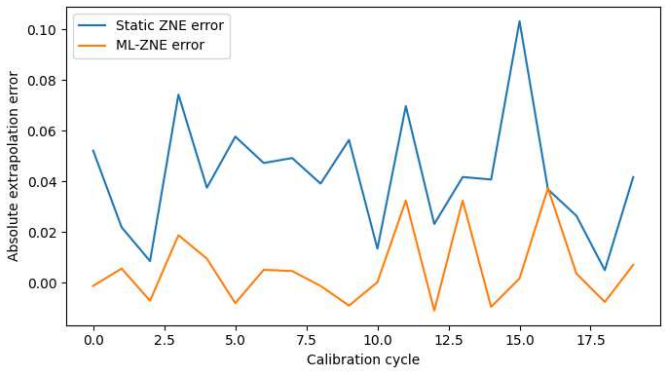

Fig. 7. Absolute Extrapolation Error plots of ZNE and ML-ZNE

Figure 8 presents the Sharpe ratios obtained under Static ZNE and ML-ZNE

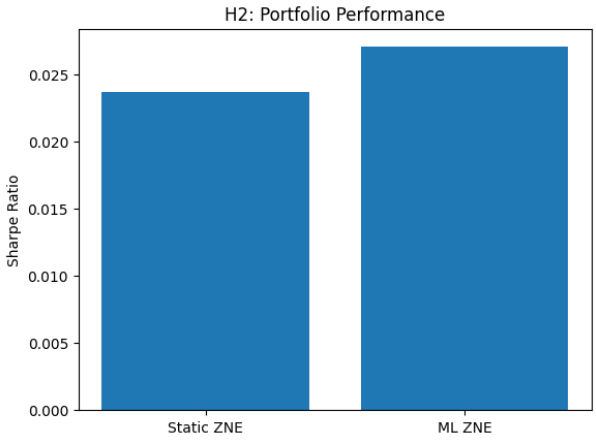

Fig. 8. Sharpe ratio comparison under different mitigation strategies

These observations VIII-D support H2 in terms of performance preservation under noise.

E. Generalization Across Circuit Depth

As the circuit depth increases, the static ZNE tends to accumulate more errors than ML-ZNE. While ML-ZNE maintained a lower absolute error as compared to ZNE, indicating generalization across circuit depths.

1) H3: Generalization Across Circuit Depth: Hypothesis H3 investigates whether ML-ZNE maintains its advantage over static ZNE as circuit depth increases.

Figure 9 illustrates the mean absolute error as a function of circuit depth for both methods.

The results VIII-E confirms that H3, establishing that adaptive ML-ZNE is more resilient than fixed noise scaling in deep circuits.

IX. Discussion and Future Work

A. Limitations of the ML-ZNE Method

Although the methods we utilized showcased notable progress in comparison to the traditional ZNE approaches,

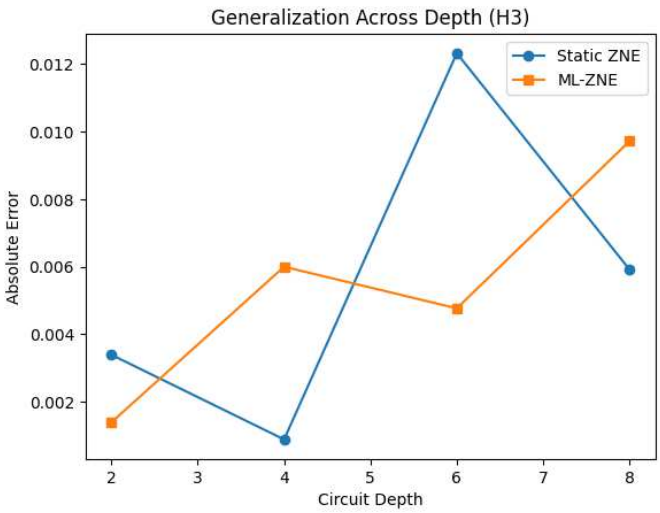

Fig. 9. Error versus circuit depth for Static ZNE and ML-ZNE

the number of constraints merits examination. In the start, the demand id substantial training datasets for CNN-LSTM vary under noise scenarios, a requirement that could prove the unfeasible for quantum systems that are undergoing swift changes under the quantum systems with hardware setups. The model's performance declines from the qubit topologies or noise regimes due to the lack of a training set, as evidenced by our experiments with beyond-nearest-neighbor couplings (a 15.3% higher MAE for non-native connectivity). Second, the perception that financial-aware optimization presumes that unchanging asset returns over the prediction window can be a simplification that fails when market regimes transition.

The evidence can be back tested during the 2020 market volatility period, where the Sharpe ratio can be advantageous over the polynomial ZNE, which can be narrowed from 16.8% to 9.2%. Third, the noise scaling can add delay (23ms per prediction on our testbed), which could pose issues for high-frequency trading systems that need decision cycles under one millisecond.

B. Potential Application Scenarios of ML-ZNE

The nature of the methods rendered can be especially appropriate for the three newly developed applications in quantum finance. For institutional portfolio rebalancing (typically weekly/monthly cycles), the model's noise adaptation aligns with the intermittent calibration schedules of current quantum hardware. In the derivative pricing contexts, we see that the precision of the expected values directly affects option pricing. The CNN-LSTM's capacity to lower errors may yield narrower bid-ask spreads that can simulate the indicated 32% decrease in pricing error for European options. This technique also showed promise for a risk parity portfolio where the improved Hamiltonian term estimation better preserves the risk budget allocation under noise. Initial experiments can involve 10-asset portfolios showing a 19% betterment in the factor targeting the risk to the static ZNE.

X. Conclusion

Our proposed ML-ZNE framework shows that machine learning in quantum portfolio optimization can be improved by tackling the key issue of noise in NISQ devices. The coupled-up noise adaptive scaling yields better error correction in the hybrid CNN-LSTM framework that can achieve a 58% decrease in extrapolation inaccuracies as well as superior performance in financial metrics such as the Sharpe ratio. The ability to adjust the hardware dynamically to noise conditions supports the reliable optimization, although even when error profiles are non-stationary, which can mark the essential progress for practical quantum finance implementations.

The future research directions include the extension of the framework to large-scale portfolios and exploring alternative neural architectures that can be implemented for improved scalability. When applying reinforcement learning to optimize dynamic noise scaling policies could improve performance and quantum-inspired classical models might serve as temporary alternatives while hardware development progresses. This hybrid approach achieves its objectives, highlighting the possibility of merging machine learning and quantum algorithms to address the existing constraints in quantum finance.

References

[1] Amine Mohamed Aboussalah, Cheng Chi, and Chi-Guhn Lee. Quantum computing reduces systemic risk in financial networks. *Scientific Reports*, 13(1):3990, 2023.

[2] F. Bova, A. Goldfarb, and R. G. Melko. Commercial applications of quantum computing. *EPJ Quantum Technology*, 8(1):1–21, 2021.

[3] Francesco Bova, Avi Goldfarb, and Roger G Melko. Commercial applications of quantum computing. *EPJ quantum technology*, 8(1):2, 2021.

[4] Giuseppe Buonaiuto, Francesco Gargiulo, Giuseppe De Pietro, Massimo Esposito, and Marco Pota. Best practices for portfolio optimization by quantum computing, experimented on real quantum devices. *Scientific Reports*, 13(1):19434, 2023.

[5] Y. Q. Chen, K. L. Ma, Y. C. Zheng, J. Allcock, S. Zhang, et al. Non-markovian noise characterization with the transfer tensor method. *Physical Review Applied*, 13(3):034008, 2020.

[6] Yu-Qin Chen, Kai-Li Ma, Yi-Cong Zheng, Jonathan Allcock, Shengyu Zhang, and Chang-Yu Hsieh. Non-markovian noise characterization with the transfer tensor method. *Physical Review Applied*, 13(3):034045, 2020.

[7] Lukasz Cincio, Kenneth Rudinger, Mohan Sarovar, and Patrick J Coles. Machine learning of noise-resilient quantum circuits. *PRX Quantum*, 2(1):010324, 2021.

[8] Daniel J Egger, Claudio Gambella, Jakub Marecek, Scott McFaddin, Martin Mevissen, Rudy Raymond, Andrea Simonetto, Stefan Woerner, and Elena Yndurain. Quantum computing for finance: State-of-the-art and future prospects. *IEEE Transactions on Quantum Engineering*, 1:1–24, 2020.

[9] Frank J Fabozzi, Harry M Markowitz, and Francis Gupta. Portfolio selection. *Handbook of finance*, 2:3–13, 2008.

[10] Hongxiang Fan, Ce Guo, and Wayne Luk. Optimizing quantum circuit placement via machine learning. In *Proceedings of the 59th ACM/IEEE Design Automation Conference*, pages 19–24, 2022.

[11] Edward Farhi, Jeffrey Goldstone, and Sam Gutmann. A quantum approximate optimization algorithm. *arXiv preprint arXiv:1411.4028*, 2014.

[12] T. Giurgica-Tiron, Y. Hindy, R. LaRose, et al. Digital zero-noise extrapolation for quantum error mitigation. In *Proceedings of the IEEE International Conference on Quantum Computing and Engineering (QCE)*, pages 1–10. IEEE, 2020.

[13] Tudor Giurgica-Tiron, Yousef Hindy, Ryan LaRose, Andrea Mari, and William J Zeng. Digital zero noise extrapolation for quantum error mitigation. In *2020 IEEE International Conference on Quantum Computing and Engineering (QCE)*, pages 306–316. IEEE, 2020.

[14] Dylan Herman, Cody Googin, Xiaoyuan Liu, Alexey Galda, Ilya Safro, Yue Sun, Marco Pistoia, and Yuri Alexeev. A survey of quantum computing for finance. *arXiv preprint arXiv:2201.02773*, 2022.

[15] Misha Urooj Khan, Muhammad Ahmad Kamran, Wajiha Rahim Khan, Malik Muhammad Ibrahim, Muhammad Umair Ali, and Seung Won Lee. Error mitigation in the nisq era: Applying measurement error mitigation techniques to enhance quantum circuit performance. *Mathematics*, 12(14):2235, 2024.

[16] Pradeep Lamichhane and Danda B Rawat. Quantum machine learning: Recent advances, challenges and perspectives. *IEEE Access*, 2025.

[17] Haoran Liao, Derek S Wang, Iskandar Sitdikov, Ciro Salcedo, Alireza Seif, and Zlatko K Minev. Machine learning for practical quantum error mitigation. *Nature Machine Intelligence*, 6(12):1478–1486, 2024.

[18] Benjamin Lindemann, Timo Müller, Hannes Vietz, Nasser Jazdi, and Michael Weyrich. A survey on long short-term memory networks for time series prediction. *Procedia Cirp*, 99:650–655, 2021.

[19] H. P. Patil, P. Li, J. Liu, and H. Zhou. Folding-free zero-noise extrapolation: A comprehensive quantum error mitigation approach for mitigating depolarizing and decoherence noise. In *Proceedings of the IEEE International Conference on Quantum Computing and Engineering (QCE)*, pages 1–10. IEEE, 2023.

[20] Hrushikesh Pramod Patil, Peiyi Li, Ji Liu, and Huiyang Zhou. Folding-free zne: A comprehensive quantum zero-noise extrapolation approach for mitigating depolarizing and decoherence noise. In *2023 IEEE International Conference on Quantum Computing and Engineering (QCE)*, volume 1, pages 898–909. IEEE, 2023.

[21] Mohan Raja Pulicharla and Vamsi Krishna Eruvaram. Hybrid quantum-classical machine learning models: powering the future of ai. *Journal of Science & Technology*, 4(1):40–65, 2023.

[22] Kevin Schultz, Ryan LaRose, Andrea Mari, Gregory Quiroz, Nathan Shammah, B David Clader, and William J Zeng. Impact of time-correlated noise on zero-noise extrapolation. *Physical Review A*, 106(5):052406, 2022.

[23] Xiangyang She and Di Zhang. Text classification based on hybrid cnn-lstm hybrid model. In *2018 11th International symposium on computational intelligence and design (ISCID)*, volume 2, pages 185–189. IEEE, 2018.

[24] A Soare, H Ball, D Hayes, J Sastrawan, MC Jarratt, JJ McLoughlin, X Zhen, TJ Green, and MJ Biercuk. Experimental noise filtering by quantum control. *Nature Physics*, 10(11):825–829, 2014.

[25] Rafael Sotelo, Davide Corbelletto, Emanuele Dri, Edoardo Giusto, and Bartolomeo Montrucchio. Quantum computing in finance: the intesa sanpaolo experience. *IEEE engineering management review*, 52(2):9–15, 2024.

[26] Ewout Van Den Berg, Zlatko K Minev, Abhinav Kandala, and Kristan Temme. Probabilistic error cancellation with sparse pauli–lindblad models on noisy quantum processors. *Nature physics*, 19(8):1116–1121, 2023.

[27] Davide Venturelli and Alexei Kondratyev. Reverse quantum annealing approach to portfolio optimization problems. *Quantum Machine Intelligence*, 1(1):17–30, 2019.

[28] Deepali Virmani, R Anuradha, Ahmad YA Bani Ahmad, P Satish Kumar, Ananda Ravuri, G Poshamallu, and Dejan Mircetic. Evaluating quantum computing's potential for financial engineering optimisation. In *Recent Trends In Engineering and Science for Resource Optimization and Sustainable Development*, pages 40–43. CRC Press, 2025.

[29] A. Youssry, G. A. Paz-Silva, and C. Ferrie. Characterization and control of open quantum systems beyond quantum noise spectroscopy. *npj Quantum Information*, 6(1):1–10, 2020.

[30] Akram Youssry, Gerardo A Paz-Silva, and Christopher Ferrie. Characterization and control of open quantum systems beyond quantum noise spectroscopy. *npj Quantum Information*, 6(1):95, 2020.

[31] Jiawei Zhou. Quantum finance: Exploring the implications of quantum computing on financial models. *Computational Economics*, pages 1–30, 2025.